Hard to Beat: Historical Rules versus Conditional Quantile Models for Peak Net-Position Limits

Abstract

Dealer peak-position forecasts improve with conditioning only in some samples and under specific benchmarks. Matched FX regression lowers loss by 14.8% and 16.6% across two retrospective periods. Against the original shorter rolling reference, earlier loss rises by 3.2%; the earlier Treasury short-history comparison rises by 144.3%. We compare matched forecasts, independent reporting systems and limit-cost rankings. Historical rules beat expanded regression in 34 of 48 additional weekly comparisons. In 17 related CFTC dealer-category futures series, F or I beats regression in 29 of 34 comparisons. Regression's median excess losses are 19.3% and 10.3% across the two histories. The 90% model confidence set retains matched historical rules in 90 of 98 comparisons, usually alongside alternatives. On fixed calibrated paths, allowing stronger penalties and I reduces short-history median losses from 116.2% and 121.4% to 0.2% and -0.2%. Parameter switching misses the registered additional-loss threshold. Limit-cost rankings change with reserve, breach-frequency and exceedance charges. Peak-limit validation therefore needs matched references, complete tuning procedures and separate error-cost components.

Keywords: Peak net positions; Quantile forecasting; Historical benchmarks; Forecast comparison; Calibration; Dealer positions.

1 Introduction

Dealer position limits concern the largest exposure reached before a review horizon ends. A position can finish within its limit after crossing it earlier (Figure 1). We compare historical and conditional forecasts of this sampled absolute-position peak.

Historical headroom combines the current position with completed past peak behavior. Conditional models estimate additional relationships from finite historical windows. Matched histories and update schedules show whether the added conditioning improves the forecast.

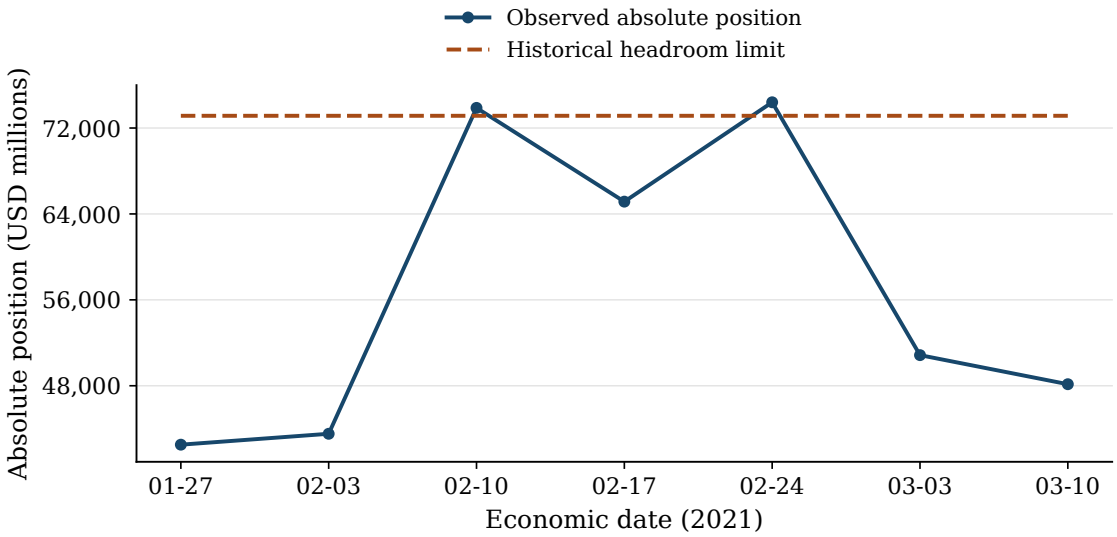


Figure 1: A selected Treasury-bill path whose endpoint returns within the historical report after an intervening breach. The rule chooses the earliest eligible original evaluation window. Endpoint containment can conceal a peak breach.

Our evidence covers hourly FX positions, their daily aggregates, New York Fed categories and CFTC dealer-category futures. The two official weekly systems report different aggregate exposures. Original protocol comparisons retain their original reference procedures and values.

Three questions organize the evaluation.

RQ1 asks whether conditional quantile methods improve sampled peak forecasts with matched histories and update schedules.

RQ2 asks whether those gains extend across frequencies and independently reported dealer-position aggregates.

RQ3 asks whether score gains imply lower limit costs under reserve, breach and exceedance preferences.

The paper makes three empirical findings.

First, matched FX gains recur, but the original-reference direction changes. Regression lowers matched loss by 14.8% and 16.6% (Table 3). Earlier loss rises by 3.2% against the original rolling reference. Delayed calibration retains the later matched gain with 329 versus 328 breaches.

Second, historical benchmarks remain competitive across reporting systems, and tuning is consequential. F or I beats expanded regression in 29/34 CFTC comparisons (Table 5). Regression's median excess loss is 19.3% at history 52 and 10.3% at 260. Matched F/I enters 90/98 model confidence sets, but is the sole method in only 2. In paired simulations, allowing stronger penalties and I changes constant/switching short-history medians to 0.2%/-0.2%. Both improve over the original menu; the registered switching increment is unsupported.

Third, lower score and lower limit cost have different preference regions. At $c = 1$, across the displayed coefficient grid, expanded regression is cheaper in 1–32 of 34 CFTC comparisons. Figure 4 retains the complete regions for all source groups. Reserve amounts, breach counts and exceedance amounts determine different parts of the ranking.

Section 3 defines the response. Section 7 answers RQ1 and RQ2; Section 8 answers RQ3. Elementary floor and endpoint properties support the forecast definitions. The forecasting, comparison and cost-accounting methods are established.

2 Related literature

Wisniewski and colleagues forecast signed next-hour position intervals using historical states, calendar information and local prediction methods (Wisniewski et al., 2020; Wisniewski, 2024). Their chronological training, calibration and test design reports scores and coverage together. The present peak response uses a complete future path segment and the current magnitude, producing different labels and limits.

Path-risk forecasts address within-period extremes. MaxVaR studies the worst cumulative portfolio return under fixed holdings and diffusion assumptions (Boudoukh et al., 2004). The payment-liquidity thesis forecasts conditional maximum deficits from confidential transactions (van den Hoogen, 2017). Settlement studies examine tail liquidity needs and collateral requirements (Petrunia et al., 2019; Arjani et al., 2020). Duffie and colleagues study dealer capacity and Treasury-market functioning, including PnL quantile forecasts (Duffie et al., 2023). These responses differ from the sampled absolute-position peak.

Regression quantiles target asymmetric forecast errors directly (Koenker & Bassett, 1978). Autoregressive quantiles and quantile forests supply other conditioning structures (Engle & Manganelli, 2004; Meinshausen, 2006). Consistent scores and elementary-score representations per-

mit comparison across evaluation preferences (Gneiting, 2011; Ehm et al., 2016). Table 1 locates the present peak response, completed labels and matched historical controls.

Table 1: Closest literature and the peak-position task. A dash marks a dimension unverified in the consulted passage. Established methods support the comparison of a common peak response.

Study	Response	Data unit	Peak floor	Label timing	Evaluation	Source scope
Wisniewski et al. (2020)	Signed endpoint interval	One FX market maker	—	Ordered train/ calibrate/ test	Scores and hit tests	One source family
Boudoukh et al. (2004)	Worst portfolio return	Stylized fixed portfolio	Different initial-value anchor	Analytic diffusion	Tail ratios and monitoring	Parameter cases
van den Hoogen (2017)	Cumulative payment-deficit peak	Confidential bank payments	Absolute-position floor —	Path simulation	EVT methodology; results partly confidential	Bank payment task
Duffie et al. (2023)	Capacity, illiquidity, PnL quantiles	Dealer/ market sources	—	Expanding PnL fits	Capacity and PnL forecasts	Several sources
Engle and Manganelli (2004)	Return quantile	Asset returns	—	Lagged predictors	Quantile scores and DQ	Several returns
Ehm et al. (2016)	Quantile/ expectile comparison	Multiple forecast examples	—	Original example designs	Elementary-score curves	Multiple tasks
Meinshausen (2006)	Conditional response quantile	Regression examples	—	Original regression splits	Conditional distributions	Several datasets
This study	Sampled absolute-position peak	FX record; weekly totals	Observed current magnitude	Completed forward labels	Scores, breaches, MCS, costs	Related aggregate tasks

The payment-thesis reading is partial. Dashes mark unverified dimensions in the consulted passages; shared source lineage leaves input-byte identity unverified.

The empirical contribution concerns matched ranking stability across responses, reporting systems and cost preferences. Established forecast tests support this comparison (Diebold & Mariano, 1995; Giacomini & White, 2006; Hansen et al., 2011). Inventory-sensitive quoting motivates exposure monitoring (Avellaneda & Stoikov, 2008); the present exercise scores passive reports on observed paths.

3 The forecast target

The target is the largest sampled absolute position from issue time through the stated horizon. Let N_t denote signed position and $\mathcal{F}_t$ the issue-time information. The peak $Y_{t,H}$, observed floor B_t and nonnegative headroom $Z_{t,H}$ satisfy the following identities.

$$Y_{t,H} = \max_{0 \le h \le H} |N_{t+h}|, \qquad B_t = |N_t|, \qquad Z_{t,H} = Y_{t,H} - B_t \ge 0.$$

A feasible limit covers the current observed exposure. Figure 1 shows an endpoint inside the historical report after an earlier breach. We score reports with pinball loss and count a strict breach when the peak exceeds the limit.

$$\rho_\tau(u) = \tau u_+ + (1-\tau)(-u)_+, \qquad I_{t,H} = \mathbf{1}\{Y_{t,H} > C_{t,H}\}.$$

Proposition 1(known-floor projection). Suppose $Y \ge B$ almost surely and the issue information contains B. Projecting any forecast c to $\max\{B, c\}$ weakly improves pinball loss for every possible outcome, with an exact gain below the floor. If q is the true conditional lower inverse quantile at level $1-\alpha$, its strict exceedance probability is at most α. At an atom of the response distribution, that probability can be smaller than the nominal tail probability.

$$\rho_\tau(Y-c) - \rho_\tau(Y - \max\{B,c\}) = \tau(B-c)_+, \qquad \Pr(Y > q \mid \mathcal{F}_t) \le \alpha.$$

Proposition 2(classical endpoint-to-peak bound). Let future increments be independent and individually symmetric about zero, with partial sums S_h. Fix a starting position x, set $B = |x|$, and let q be the population $(1-\alpha/2)$ quantile of $|S_H|$. The triangle inequality and the classical scalar Lévy maximal inequality then imply the following bound, including strict inequalities at atoms (Sussman, 2023). Neither identical increment distributions nor finite increment moments are required for this population statement.

$$\Pr\left(\max_{0 \le h \le H} |x + S_h| > B + q\right) \le \Pr\left(\max_h |S_h| > q\right) \le 2\Pr(|S_H| > q) \le \alpha.$$

The Supplement proves both statements and compares population quantiles with their empirical plug-ins. A conditional application requires independence, symmetry and the stated endpoint quantile under the law given issue information. The separate bound simulations distinguish mean-reverting levels from serially correlated increments.

Separately fitted horizon forecasts can violate the ordering of true peak quantiles. Short-history FX regression crosses across horizons on 640 evaluation origins. Label nesting, floor coverage and strict-breach definitions provide additional checks on the issued reports.

4 Data

The FX application uses the hourly Net Position Time Series Dataset, reporting market-maker client-flow positions (Lindsay, 2020; Wisniewski et al., 2020). It contains 18,576 records in source USD units. The later, earlier and full-prefix comparisons use distinct chronological issue grids and the same sampled-peak definition.

The weekly data are New York Fed Primary Dealer Statistics (Federal Reserve Bank of New York, n.d.). Reporting definitions follow the government-securities dealer framework and position-accounting instructions (Board of Governors of the Federal Reserve System, 2013, 2024). The Treasury-bill series has 706 economic-date observations in USD millions. The values are fair-value positions aggregated as long-minus-short totals. Additional categories pass metadata and calendar checks and use complete weekly grids, with separate fits by category.

Daily aggregation takes the last recorded position on each nonempty source-naive calendar day. It retains partial boundary days, yielding 930 days and 501 common evaluation issues. Reports

use the completed day's end as their issue clock. The resulting target is a peak over subsequent daily records.

The CFTC application is a retrospective survivor-universe, economic-date replay. It uses futures-only Dealer/Intermediary long positions minus short positions (Commodity Futures Trading Commission, n.d.). The fixed screen requires uninterrupted report weeks from 2010, stable literal codes, exchange and units. It excludes documented consolidation-code families before ranking eligible contracts by full-sample mean open interest. Seventeen contracts qualify, fewer than the planned maximum of twenty. The Supplement lists every selected code and exclusion reason.

There are 877 weekly positions through 20 October 2026. Validation uses 52 origins; evaluation uses the last 260 eligible origins with horizon six. The evaluation position dates run from 21 September 2021 to 8 September 2026. Its final peak label completes on 20 October 2026. Refit frequency is thirteen records; histories are 52 and 260. Positions usually refer to Tuesday and are released Friday; this application replays downloaded economic dates.

Every method within a comparison uses identical peak labels, current floors and issue origins. Source units and histories remain distinct across series. The same-record aggregation holds dealer identity fixed while changing the measurement clock.

5 Forecasting methods

Forecasts target the conditional quantile at $\tau = 19/20$. The horizons $H \in \{6, 24\}$ count subsequent source records, and each peak includes the issue position. Calendar gaps extend an FX horizon's elapsed span; weekly and daily steps represent different review intervals. The nineteen-to-one error-weight preference defines the evaluation score.

Historical headroom uses the corrected empirical rank of completed headrooms and adds its report to the current magnitude. Translated paths reanchor historical signed paths, while historical-peak quantiles are projected to the observed floor. The endpoint plug-in uses an absolute-displacement quantile; Table 2 gives the executed menus and schedules.

Ridge quantile regression links mature headroom to issue-known position states and calendar variables. The larger set contains 33 predictors plus an unpenalized intercept. The four-predictor variant uses current magnitude, signed level and two increment RMS measures. All standardization uses mature training observations. Expanded-menu full regression changes only the penalty grid.

Local quantiles use neighboring historical states, with an increment-scaled variant. Filtered paths simulate positions from GARCH-standardized empirical innovations drawn independently with replacement (Hull & White, 1998; Barone-Adesi et al., 1999). Quantile forests use leaf distributions; tail methods fit volatility-scaled headroom (Meinshausen, 2006; McNeil & Frey, 2000). The Supplement lists complete results and numerical or structural failures.

Further comparisons apply delayed calibration, fixed combinations and conservative selection to the same peak response. A headroom CAViaR recursion and calibration wrappers follow existing methods (Engle & Manganelli, 2004; Gibbs & Candès, 2021). Calibration parameters update after outcomes mature. Fixed mean and median combinations use predefined members. Conservative selection favors the strongest shrinkage within one paired HAC standard error of the validation winner. This adapts one-standard-error selection (Hastie et al., 2009) to paired forecast losses.

Table 2: Forecast definitions and executed menus. Codes occur here to connect with the result files. Matched histories and update schedules make the reference explicit.

Method / code	Forecast definition	Executed menu / setting
Historical headroom (F)	Current floor plus a quantile of completed peak headrooms	FX histories {720,2880}; weekly {52,260}
Translated path (S)	Translate completed signed paths to the current position before taking their peaks	Same completed-label pools; fixed-window and rolling-window controls
Historical peak (K)	Historical absolute-peak quantile projected to the current floor	Same histories and update controls
Quantile regression (QR)	Training-scaled headroom regression with ridge slopes and an unpenalized intercept	Original three-point menu; six-point menu $\{0.001, 0.01, 0.1, 1, 10, \infty\}$
Local quantiles (KNN)	Neighbor headroom quantile, with a separate increment-scale variant	Training-only transforms and fixed neighbor menus; Supplement
Parsimonious regression (QR4)	Current signed/magnitude positions and short/long increment RMS	Original five-point menu; six-point paired comparison below
Endpoint plug-in (L)	Floor plus a high absolute-displacement endpoint quantile	Proposition 2; empirical plug-in evaluated separately
Further benchmarks	Autoregressive replay; filtered paths; peak-tail fit; quantile forest	Existing bounded specifications, complete failures included in Supplement
Coverage and combinations	Delayed projected ACI-style wrappers; headroom SAV; fixed forecast combinations	Original-source wrapper/SAV inventory and broader combination inventory are separate

Training histories count completed labels. Fixed-window models refit every 24 FX records or 13 weekly records. Quantile and normalization conventions are detailed in the Supplement; this table does not expand any executed menu.

6 Evaluation design

At refit r, training labels are eligible only when $i + H \leq r$. Predictors use issue-known observations, and training transforms remain fixed until the next refit. Original comparisons select parameters in the preceding validation segment. The full-prefix backcast holds its specified penalties fixed. Weekly selection uses 52 outer validation origins with overlapping labels.

Matched comparisons fix completed-response history and refit schedule for regression and historical headroom. The main FX reference uses a shorter rolling history. The main Treasury reference shares regression's longer history and refit schedule. Tables 3 and 4 report both alongside the other comparison sets.

Accuracy uses mean pinball loss; breaches are strict peak exceedances. Evaluation scores passive reports on the observed position paths. Paired loss differences use HAC covariance with the horizon-based bandwidth rule (Diebold & Mariano, 1995; Newey & West, 1987). Tables identify the small-sample adjustment and named multiplicity families (Harvey et al., 1997; Holm, 1979). Hit diagnostics use a fixed spaced grid and condition the transition likelihood on its initial hit (Kupiec, 1995; Christoffersen, 1998).

6.1 Research design and transparency

The original protocol fixes the two main comparisons and their reference procedures. Other p-values are exploratory and unadjusted unless stated. The Supplement records comparison families, execution chronology and complete failure inventories. The new studies retain separate estimation, simulation and reporting records. Menu design follows the earlier simulation results but precedes paired computations. CFTC design precedes acquisition; two format clarifications precede all forecasts. MCS and cost specifications lock before calculations on previously inspected reports. These are frozen descriptive decision rules within a retrospective research program.

6.2 Multiple-method forecast comparison

We form a 90% model confidence set separately for each complete source, period, horizon and history grid (Hansen et al., 2011). The range statistic uses 1,000 stationary-bootstrap replications in arch 8.0.0. Block length is $\max\{H, \lceil n^{1/3} \rceil\}$. All complete saved procedures enter; incomplete prior methods remain explicitly unavailable. Entire origin grids and targets must match. The nominal level presumes suitable loss dependence; it does not adjust earlier research-stage model development.

Exact duplicate loss columns enter the bootstrap once and their procedure labels are unfolded afterward. Matched F/I aliases merge only after full forecast and target equality. Rolling references remain distinct. Literal sole membership differs from a sole surviving loss class containing several procedures. The fractions summarize related tasks, with no independent cross-market pooled test.

7 Results

7.1 RQ1: matched accuracy and reference stability

Long-history FX regression beats matched historical headroom in both periods. Loss falls by 14.8% later and 16.6% earlier. Table 3 separates these matched controls from the shorter rolling reference.

Delayed calibration retains the later matched FX gain. Wrapped regression lowers loss by 15.0%. Its 329 breaches compare with 328 for wrapped historical headroom on common origins (Table 3).

The Supplement reports the training-defined state, target-level and overlapping full-prefix comparisons. Full-prefix regression lowers matched loss by 12.2% but records 416 versus 282 breaches. Its rolling-reference DM p-value is 0.055. Longer-horizon gains coexist with reference-sensitive losses in the complete tables.

Table 3: Long-history short-horizon FX regression under two references and periods. Negative percentages favor regression. Matched gains recur while the rolling-reference direction changes.

Period / procedure	Reference	Loss change (%)	Issues	Strict breaches (regression / reference)
Later / original main comparison	Shorter-history rolling headroom	-12.66	6625	376 / 332
Later / matched	Same-history fixed-window headroom	-14.77	6625	376 / 360
Earlier / original structure	Shorter-history rolling headroom	+3.17	1440	See complete earlier-period tables
Earlier / matched	Same-history fixed-window headroom	-16.62	1440	20 / 16
Later / coverage-wrapped	Same-history wrapped headroom	-15.00	6625	329 / 328

Regression uses history 2880 and horizon 6; the original rolling reference uses 720. The earlier matched comparison is the original three-penalty regression, not the later one-standard-error selection rule. The wrapper comparison is a separate finite projected delayed ACI-style exercise.

7.2 RQ2: daily, weekly and independently reported positions

The earlier FX ranking reverses when the reference changes. Against shorter-history rolling headroom, later loss falls by 12.7%, but earlier loss rises by 3.2%. Matched gains and rolling-reference gains therefore have different period stability (Table 3).

Short-history FX regression performs poorly at the longer horizon. Loss rises by 31.5% against matched historical headroom. Regression records 924 breaches against 427 for the reference; the full horizon map retains the other adverse comparisons.

Treasury-bill regression loses on both accuracy and breaches. Mean loss rises by 50.8%, with 54 regression breaches. Historical headroom records 28 breaches; the earlier short-window version also loses (Table 4).

Stronger regularization narrows weekly panel losses but leaves historical rules ahead. Median excess loss falls from 92.6% to 9.2% after menu expansion. Regression still loses 34 of 48 matched comparisons. Conservative selection chooses I in 17 of 26 earlier weekly cells but still leaves

a worst-cell increase of 181.4%. This is the Treasury-bill short-window, 24-step comparison (Table 4).

Simpler conditional reports and fixed combinations also give mixed weekly results. The median combination lowers Treasury-bill loss by 4.0%, while the mean combination raises it by 2.8%. The headroom recursion is also worse. The Supplement retains every completed loss and model failure.

Historical headroom wins seven of eight same-record daily comparisons. The exception uses four predictors, history 260 and horizon 6, reducing loss by 3.0%. It records 37 breaches versus 30 for matched headroom. Daily aggregation therefore preserves the score–breach disagreement while changing the leading forecast in most comparisons.

Table 4: Weekly evidence across the original main comparison, added panel and earlier-period comparisons. Counts refer to related tasks rather than independent dealers. Stronger regularization narrows losses while leaving substantial failures.

Comparison set	Procedure	Loss change / median (%)	Higher-loss cells
Original Treasury-bill main comparison	Original regression, long history	+50.79	Original main comparison
Earlier Treasury-bill short-window version	Short-history short-window version	+144.30	Different history
Additional panel	Original full regression	+92.63	45 / 48
Additional panel	Four-predictor regression	+9.91	37 / 48
Additional panel	Full regression, expanded penalties	+9.22	34 / 48
Additional panel	Ordinary empirical headroom quantile	+0.50	30 / 48
Earlier weekly groups	Validation-minimum expanded rule	+4.15	26 groups
Earlier weekly groups	One-standard-error rule	+5.37	26 groups

The original bill main comparison has 54 versus 28 strict breaches among 277 issues. Earlier short-history bill results are a short-window version, not an exact replay of that main comparison. The one-standard-error worst excess is +181.39%; all its prewritten success criteria remain unmet. The Supplement lists all panel and selection results.

7.2.1 CFTC: independent reporting-system evidence

F or I has lower loss than expanded-menu regression in 29 of 34 contract/history comparisons. The frozen sixty-percent threshold is met. The count uses the better realized score among the two prespecified historical reports, not an executable F/I selector. F remains the fixed matched reference; I is reported separately. Regression's median excess losses are 19.3% and 10.3% at histories 52 and 260. Both exceed the registered ten-percent tolerance. Four-predictor regression improves from 10.3% to 0.2%. Thus the historical-benchmark pattern extends to another reporting system, with substantial contract-level variation.

7.2.2 Model confidence sets

Matched F/I remains in 90/98 sets, while 197/683 complete conditional-procedure slots are excluded. Only 2 sets contain a single literal historical method. These results support retaining historical controls while leaving many alternatives uneliminated under the fixed bootstrap approximation. Non-elimination does not establish equal predictive risk. Table 6 distinguishes

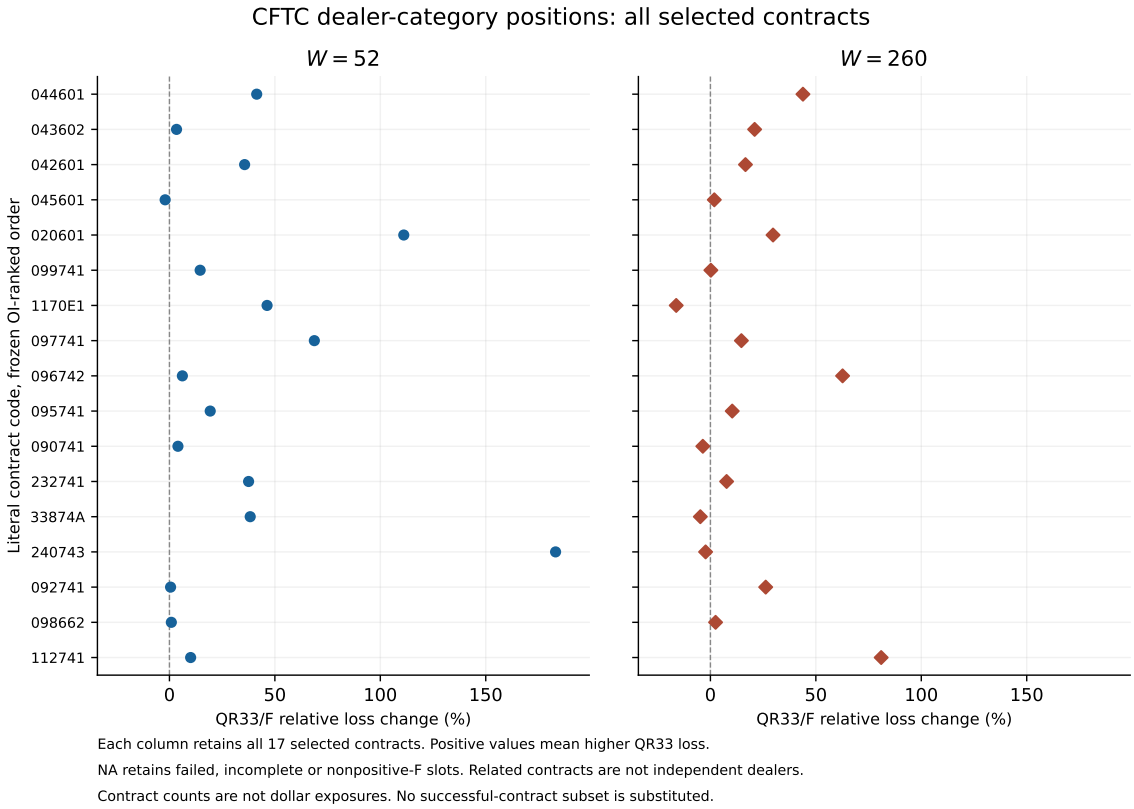


Figure 2: Expanded-menu regression relative loss versus matched F for every eligible CFTC contract. Columns separate histories; code order follows the fixed open-interest ranking. Positive values mean higher regression loss.

Table 5: CFTC dealer-category futures application: within-contract relative loss versus matched F (%). All 17 selected literal contract codes are retained.

Contract code	$W = 52$					$W = 260$				
	I	QR4	QR33	ACI-F	ACI-QR33	I	QR4	QR33	ACI-F	ACI-QR33
044601	5.26	128.00	41.38	15.01	18.80	0.51	7.82	43.88	13.87	42.41
043602	3.37	9.35	3.37	37.79	24.64	2.30	8.95	20.97	-11.22	-12.64
042601	4.64	9.48	35.65	26.69	41.21	-0.15	74.23	16.63	1.46	9.39
045601	-2.04	0.31	-2.04	21.83	6.14	1.85	1.85	1.85	2.40	2.58
020601	4.77	79.72	111.12	25.66	-5.34	0.92	43.38	29.70	-15.12	-12.32
099741	0.69	13.97	14.56	77.33	-13.28	0.22	0.22	0.22	27.72	27.73
1170E1	6.42	6.82	46.30	-0.22	12.77	0.92	-28.44	-16.22	-1.12	1.00
097741	-0.83	23.76	68.71	29.77	33.79	0.53	-6.54	14.69	-0.41	-5.43
096742	2.93	3.41	6.13	49.54	30.46	5.30	4.82	62.68	37.12	-0.49
095741	7.39	19.55	19.33	41.27	42.98	-0.70	2.13	10.35	19.30	55.39
090741	4.03	10.28	4.03	15.58	13.13	3.28	-5.35	-3.60	-2.02	-5.66
232741	0.11	123.33	37.53	63.84	36.35	1.94	-3.35	7.68	-0.65	-3.86
33874A	6.95	88.57	38.30	30.15	44.07	-2.38	-3.21	-4.78	6.98	9.52
240743	-3.04	-7.32	183.12	33.14	49.77	-2.28	-2.28	-2.28	32.91	26.37
092741	0.54	0.54	0.54	-1.74	-0.68	0.17	-6.98	26.22	5.58	13.94
098662	-0.44	-0.44	0.90	44.89	38.90	2.45	-39.44	2.45	15.12	20.45
112741	11.06	38.76	10.05	33.72	24.10	2.24	2.82	80.98	8.36	16.37

Each entry is $100(\overline{L}_m/\overline{L}_F - 1)$ using the saved complete evaluation. NA retains failed/incomplete methods or an unavailable/nonpositive F denominator. Positive values indicate higher loss. T1 uses every selected-contract/two-window cell (denominator 34), with the registered fraction at least 0.60; T2 requires the all-contract QR33 median at both windows inside [-10.00, 10.00]%. T3 compares complete all-contract QR4 medians strictly. No successful-contract subset substitutes for any threshold. Displays are rounded; decisions use saved full-precision operands. F and every absolute score/breach/selection receipt remain in the detailed summary and exported cell CSV.

literal membership from exact observed-loss equality. Every set, method p-value, equivalence class and prior unavailable method appears in the Supplement.

7.3 Scores and breach behavior answer different questions

Lower FX loss coexists with more breaches before calibration. Regression breaches 376 times against 360 for matched historical headroom. Pinball loss measures error amounts, while a strict-hit indicator records event occurrence. Monitoring reports should show both.

Elementary-score rankings cross for both original main comparisons (Supplementary Murphy diagrams). A separate 24-step FX comparison against rolling translated paths reverses near 20.6:1. Both six-step FX comparisons improve mean overrun and unused allowance. The weight reversal therefore belongs to the 24-step reference.

Loss and hit diagnostics answer different questions (the Supplementary diagnostics). HAC tests compare average loss; strict-hit tests examine event frequency and serial behavior. RMS-state diagnostics compare conditional predictive ability across training-defined groups (Giacomini & White, 2006).

Table 6: Ninety-percent model confidence sets across all planned comparisons. Counts retain their own comparison or conditional-procedure denominators.

Group	Complete / planned	Matched F/I included	Sole literal F/I	Conditional excluded	Prior NA methods
FX original	4/4	2/4 (50.0%)	0/4 (0.0%)	39/66 (59.1%)	6
FX early2015	4/4	3/4 (75.0%)	0/4 (0.0%)	1/15 (6.7%)	2
Treasury bills	4/4	4/4 (100.0%)	0/4 (0.0%)	11/74 (14.9%)	4
Other NYFed panels	48/48	46/48 (95.8%)	0/48 (0.0%)	110/384 (28.6%)	48
Same-record dailyFX	4/4	4/4 (100.0%)	2/4 (50.0%)	5/8 (62.5%)	0
CFTC	34/34	31/34 (91.2%)	0/34 (0.0%)	31/136 (22.8%)	0
All comparisons	98/98	90/98 (91.8%)	2/98 (2.0%)	197/683 (28.8%)	60

All planned comparisons complete, so successful and planned denominators coincide. Matched F/I excludes rolling references and query-nearest state baselines. Conditional exclusions count complete procedure slots; prior NA methods are not statistical exclusions. A sole exact-loss class can retain several procedures: 4/98 sets have such a class containing F/I, versus 2/98 sole literal historical methods. Related tasks are not independent replications. The Supplement retains every set, p-value, loss class, secondary membership measure and availability record.

7.4 Isolating the penalty menu

Earlier stationary comparisons already produce large short-history regression losses. The fitted constant and switching arms use thirteen pre-evaluation weekly calibrations. Their original history-52 medians are 116.2% and 121.4%. The earlier stability screen rejects seven series pointwise and two after Holm correction. The Supplement retains its diagnostics and all earlier simulation grids.

The paired menu experiment changes only the ridge candidate menu and the stated exact-tie rule. It adds penalties one, ten and the intercept-only limit to full regression. Both regressions use the six-point menu. All paths, innovations, source calibrations and forecasting clocks match the earlier experiment. Thirty-two paired repetitions contribute to each complete source median. This isolates the executed menu comparison within these calibrated laws. The expanded procedure can choose ordinary I. Stronger finite penalties and access to I are not separately identified; near-parity loss alone supplies no evidence of added feature value.

At history 52, the expanded full-regression medians are 0.2% under constant parameters and -0.2% under switching. Both arms improve in all thirteen sources. The constant-arm near-parity prediction and the paired-improvement prediction hold. The switching-minus-constant difference misses the registered five-percentage-point increase. This outcome supports the menu explanation for these simulated losses; it leaves the empirical frequency mechanism unresolved.

At history 260, one required full-regression repetition fails in each arm under the old low penalty. Both whole thirteen-source summaries remain unavailable. The experiment retains every failed cell and performs no retry. Earlier daily moment-qualification and hourly optimization failures also remain unavailable. Absence of a five-point switching increment is not evidence of equivalence.

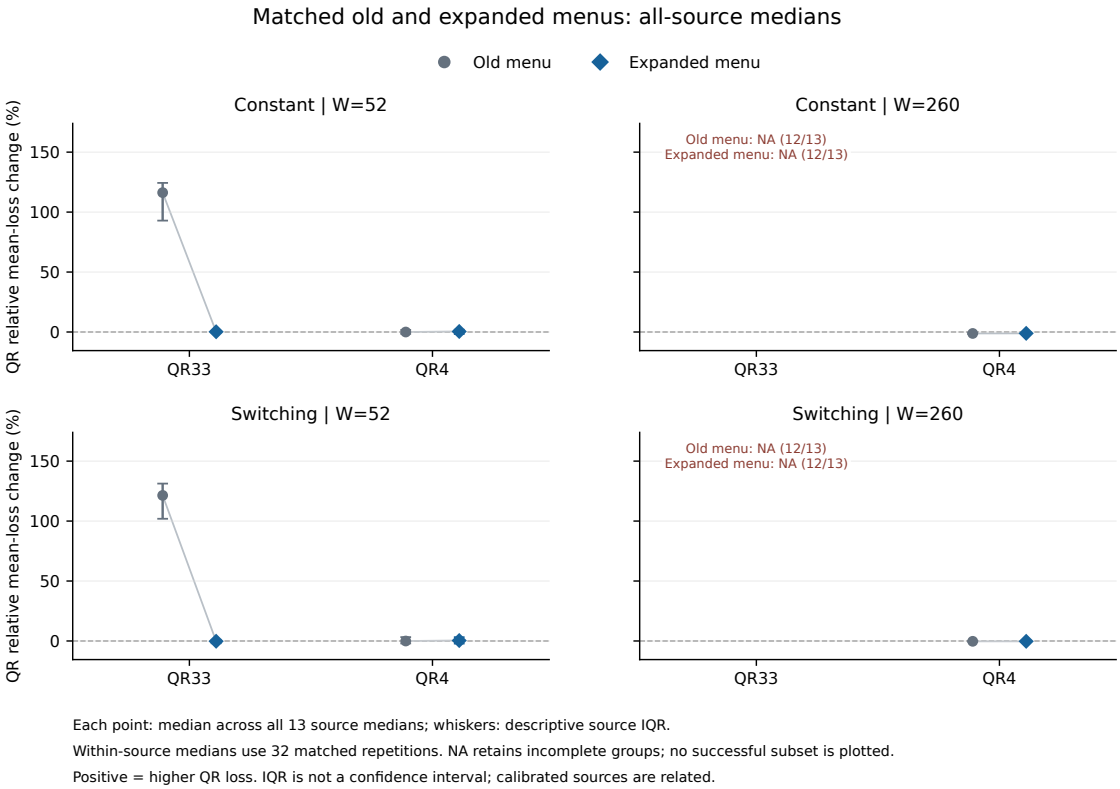


Figure 3: Paired penalty-menu isolation on unchanged calibrated paths. The expanded menu removes the large short-history full-regression loss in both arms. Longer-history full-regression summaries remain unavailable because one required source repetition fails in each arm. No successful subset replaces them.

Table 7: Penalty-menu isolation on matched calibrated paths. Entries are the median relative loss (%) across all 13 source medians, with the descriptive source IQR in brackets.

Arm	W	Method	Old menu Median [IQR]	Expanded menu Median [IQR]	Missing sources Old / expanded
Constant	52	QR33	116.24 [92.90, 124.34]	0.21 [-0.52, 0.98]	0 / 0
Constant	52	QR4	-0.02 [-1.55, 1.73]	0.44 [-1.69, 1.90]	0 / 0
Constant	260	QR33	NA	NA	1 / 1
Constant	260	QR4	-1.21 [-1.69, -0.54]	-1.13 [-1.69, -0.64]	0 / 0
Switching	52	QR33	121.40 [101.92, 131.26]	-0.23 [-1.88, 0.98]	0 / 0
Switching	52	QR4	-0.03 [-2.03, 3.22]	0.38 [-2.17, 3.17]	0 / 0
Switching	260	QR33	NA	NA	1 / 1
Switching	260	QR4	-0.27 [-0.66, 0.23]	-0.28 [-0.75, 0.48]	0 / 0

Each source median uses all 32 matched repetitions. Relative loss is $100(\overline{L}_{QR}/\overline{L}_F - 1)$ within each repetition. An unavailable old or expanded all-source aggregate is shown as NA; successful subsets are not substituted. Positive values mean higher QR loss. Displays are rounded; registered decisions use the saved full-precision operands. The expanded QR4 exact-tie rule prefers stronger penalties, unlike its old rule. Saved exact ties: 0; changed QR4 tie selections: 0.

8 RQ3: limit-cost regions

Let $m_F = \overline{C_F - B} > 0$ be matched F's average reserved headroom in a comparison. We evaluate each saved report using

$$J_m(\kappa, \beta, c) = \kappa \frac{\overline{C_m - B}}{m_F} + \beta\, \overline{\mathbf{1}\{Y > C_m\}} + c \frac{\overline{(Y - C_m)_+}}{m_F}, \qquad \beta = b/m_F.$$

The fixed breach amount is $b = \beta m_F$. Reserve, breach frequency and excess enter separately. Costs apply per issued report; overlapping horizons are not added as separate cash flows.

The fixed grid uses $\kappa \in \{.01, .025, .05, .1, .2, .5, 1\}$, $\beta \in \{0, .01, .05, .1, .25, .5, 1, 2, 5\}$ and $c \in \{.5, 1, 2\}$. Every complete saved method enters all grid points. Each beta fixes breach charges relative to that cell's evaluation-sample F headroom. Monetary charges differ across cells. The panels compare fixed reports, not a cost-optimal policy frontier. Figure 4 fixes $c = 1$ and the expanded-menu full regression for hourly and weekly sources. The daily comparison retains its saved original-menu regression. No method is selected for having the lowest cost.

The cheaper-method region changes with the cost coefficients in every source group. In CFTC, expanded regression wins 1–32 of 34 comparisons across the displayed grid. High reserve charges favor lower issued headroom; larger fixed breach charges shrink those regions. The Supplement and zipped table retain all methods and coefficient combinations.

At $\kappa = .05, \beta = 0, c = 1$, the cost difference from F equals its pinball-loss difference divided by m_F. Other cost weights need not preserve that ranking. Breach frequency contributes information beyond the two error-amount components.

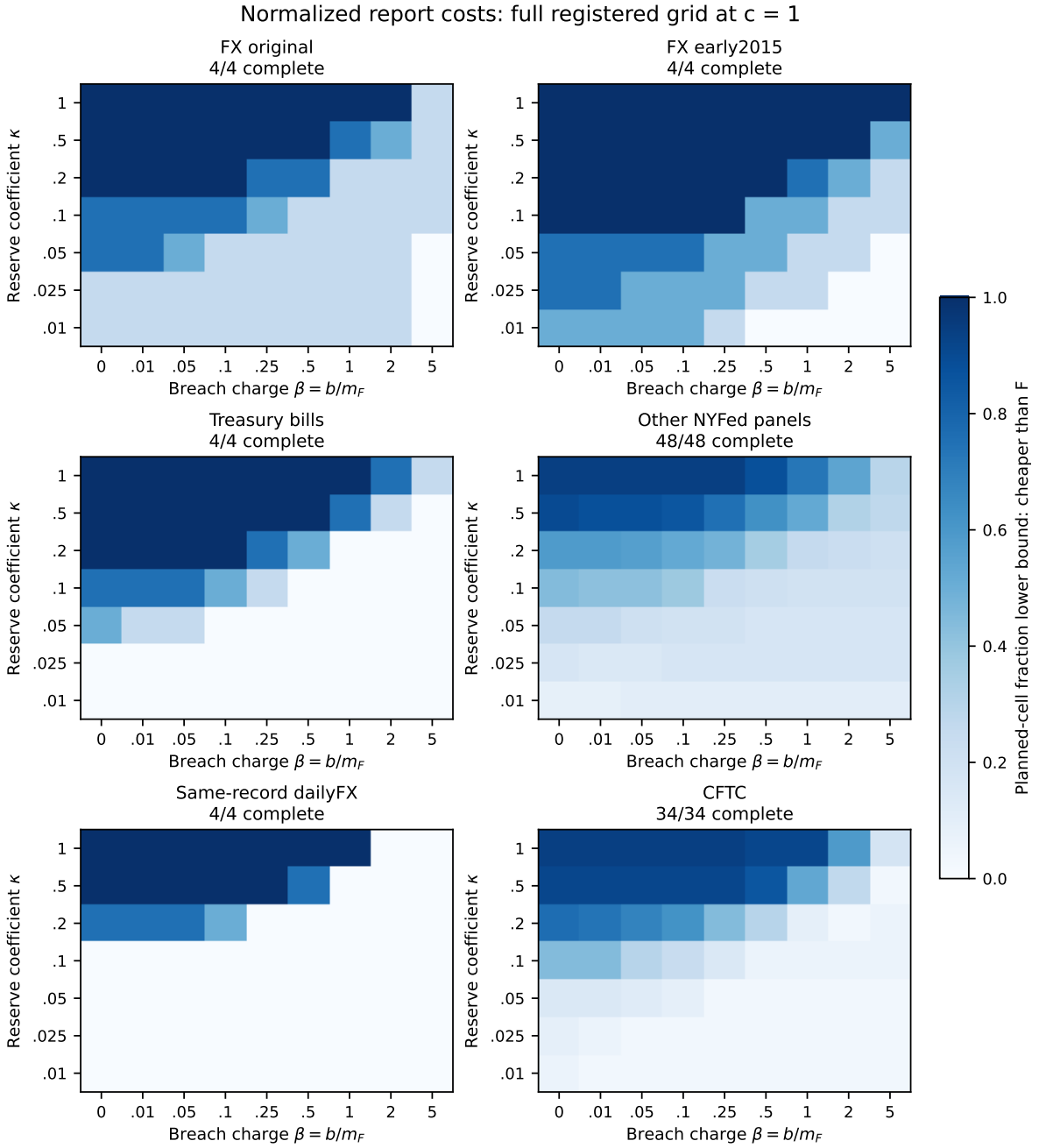


Figure 4: Complete registered reserve/fixed-breach regions at $c = 1$. Color is the planned-comparison fraction where the fixed conditional procedure costs less than F. Each panel retains its source, horizon and history grids with equal comparison weights. These fractions are not probability estimates. All comparisons complete, so lower and upper fractions coincide.

9 Implications for model validation

Benchmark against historical headroom. Use the same completed labels, history length and refit schedule. Name the shorter rolling reference separately, since its FX ranking changes across periods.

Retain corrected-rank history as a matched benchmark in these comparisons. Expanded penalties reduce losses, while short-sample validation, including one-standard-error selection, still leaves large losses. The CFTC comparison also retains F/I in most model confidence sets. The ordinary empirical quantile is up to 36.5% worse than corrected headroom in the earlier weekly grid.

Report breaches with the score. Show mean underprediction, mean unused headroom and strict-breach count on the same grid. The unwrapped matched FX gain records 376 versus 360 breaches; the daily regression exception also has more breaches. Cost regions add reserve, breach-frequency and exceedance preferences to this joint evaluation.

10 Limitations

The FX evidence describes a single market maker's client-flow record. Its original evaluation period was viewed before the conditional methods were specified. Rules for the earlier 2015 comparison followed the original results. The full-prefix backcast uses penalties selected after its issue dates and contains the earlier backcast. Its target paths were covered by previous scoring. Both exercises remain retrospective.

The New York Fed weekly categories share a reporting system and economic components; they are related aggregate tasks, not independent dealer replications. Treasury comparisons replay the downloaded economic-date vintage. Historical publication-time information is not reconstructed. Aggregation can offset gross dealer exposure; sampled net inventory also differs from settlement cash needs. Valuation and reporting-composition changes are not separated from trading. Sampled peaks omit intervening excursions. Daily last records also differ from verified closes or intraday maxima; the source timezone and release vintage are unknown.

Daily aggregation jointly changes physical histories, horizons, update cadence, calendar predictors, evaluation span and sampled targets. The same-record comparison leaves a pure frequency effect unidentified.

The design gives no causal interpretation or individual-predictor effects. The financial procedures provide no coverage guarantee under their dependence and calibration scheme. Trading and funding gains are not measured. Scoring weights are evaluation preferences, not estimated intervention or funding costs. Independent dealer records and prospective observations would broaden validation; finer exposure and cost data would connect reports to inventory actions.

Short overlapping validation samples and research-stage selection limit tuning stability and confirmatory inference. Longer weekly outer validation windows remain untested. The simulation designs use different menus, clocks, calibration laws and summary statistics. The paired menu experiment isolates that procedure change within the calibrated laws; empirical causation remains unresolved. The new parameter-switching process repeats fitted dwell lengths and adds artificial last-to-first transitions. Its stability screen uses a plug-in model-based approximation. Moment qualification and successful optimization leave the assumed law and boundary-estimate uncertainty unresolved. Failed cells and unavailable diagnostic comparisons retain those statuses.

CFTC categories provide an independent reporting system, not independent individual dealers. Full-history continuity creates a retrospective survivor universe. Open-interest ranking only

orders the eligible contracts here. Conservative consolidation-family exclusions leave seventeen eligible contracts. Raw contracts have different exposure units; normalized comparisons do not recover common-dollar risks. Downloaded position dates omit actual historical first-release vintages and reporting delays. Two ingestion-schema clarifications followed acquisition but preceded every forecast and model-loss result; their original failed technical attempts remain documented.

The limit costs are illustrative, not observed real operating costs. Coefficients are fixed preferences; no estimated funding return, optimal intervention policy or trading-profit claim follows. The normalization uses evaluation-sample F headroom; ex ante institutional charges remain untested. MCS inference requires suitable loss dependence. The stationary bootstrap, modest samples and research-stage specification choices limit its nominal interpretation. Membership and cross-task fractions do not certify future coverage or simultaneous best-model status.

11 Conclusion

RQ1 has a reference-dependent answer. Hourly FX regression lowers matched loss by 14.8% and 16.6% across periods. Delayed calibration retains the later gain with 329 versus 328 breaches. Earlier loss is 3.2% higher against the original shorter rolling reference; that direction fails to recur. The earlier Treasury short-history comparison raises loss by 144.3%.

For RQ2, historical rules remain strong across related weekly categories, same-record daily aggregates and the independent CFTC reporting system. F or I beats expanded regression in 29/34 CFTC comparisons. Regression's median excess losses are 19.3%/10.3% at the two histories. Matched F/I survives in 90/98 model confidence sets, usually alongside other methods. Expanded menus reduce weekly losses, but short-sample validation still fails to protect the historical benchmark consistently. With stronger penalties and I, paired simulations reduce constant/switching short-history medians to 0.2%/-0.2%; the registered switching increment is unsupported. The empirical low-frequency mechanism remains unidentified.

For RQ3, score gains translate into lower limit costs only in parts of the coefficient grid. At $c = 1$, CFTC regression is cheaper in 1–32 of 34 comparisons as reserve and breach charges vary. Limit validation should match histories and schedules, retain historical controls, and report reserve amounts, breach counts and excess amounts together.

Data and code availability

Data availability. The FX input is the specified version of the Kaggle Net Position Time Series Dataset. The owning metadata labels its license "Data files © Original Authors." This repository supplies the acquisition route and input fingerprint, but excludes raw FX observations and private per-issue prediction vectors. Obtain the source from `https://www.kaggle.com/datasets/davidlindsay1979/net-position-time-series-dataset` subject to the original rights holders' terms.

The New York Fed aggregates are available from its Primary Dealer Statistics service under the Bank's Terms of Use. The repository preserves the unchanged accepted Treasury snapshot with attribution and identifies study calculations separately. Copying or distributing that content remains subject to its source notices, attribution, modification labelling, continuing conditions and endorsement restrictions: `https://www.newyorkfed.org/privacy/termsofuse.html`. The code license does not replace those source conditions.

The CFTC application uses the public TFF historical files from `https://www.cftc.gov/MarketReports/CommitmentsofTraders/HistoricalCompressed/index.htm`. The repository

includes unchanged downloaded archives and a per-file URL, acquisition time, byte count and SHA-256 record in `data/cftc-tff/provenance.json`.

Code availability. Code, saved results and version-specific reproduction instructions are available at `https://github.com/ByronLi0207/financial-mathematics-paper-20261020/tree/revision-v10`. The v10 package adds paired menu isolation, official CFTC sources, full MCS results and fixed cost regions. Earlier results remain unchanged in their versioned directories. Document compilation and restricted-input financial replay have separate instructions. Complete, adverse, failed and unavailable outcomes are retained.

Author-provided information

Author information: [TO BE COMPLETED BY AUTHOR].

ORCID: [TO BE COMPLETED BY AUTHOR].

Funding: [TO BE COMPLETED BY AUTHOR].

Competing interests: [TO BE COMPLETED BY AUTHOR].

AI statement: [TO BE COMPLETED BY AUTHOR].

References

Arjani, N., Li, F., & Sabetti, L. (2020). *Monitoring intraday liquidity risks in a real time gross settlement system*. Payments Canada. `https://www.payments.ca/sites/default/files/discussion_paper_monitoring_intraday_liquidity_risks_in_a_real_time_gross_settlement_system.pdf`.

Avellaneda, M., & Stoikov, S. (2008). High-frequency trading in a limit order book. *Quantitative Finance*, *8*(3), 217–224. `https://doi.org/10.1080/14697680701381228`.

Barone-Adesi, G., Giannopoulos, K., & Vosper, L. (1999). VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets*, *19*(5), 583–602. `https://doi.org/10.1002/(SICI)1096-9934(199908)19:5<583::AID-FUT5>3.0.CO;2-S`.

Board of Governors of the Federal Reserve System (2013, March). *Reporting guidelines for preparing the FR 2004 primary government securities dealers reports*. Board of Governors of the Federal Reserve System. `https://www.federalreserve.gov/apps/reportingforms/Download/DownloadAttachment?guid=f555fe11-e149-479b-8088-4ed4e9ae1206`.

Board of Governors of the Federal Reserve System (2024, July). *Instructions for the preparation of government securities dealers reports: Reporting form FR 2004*. Board of Governors of the Federal Reserve System. `https://www.federalreserve.gov/apps/reportingforms/Download/DownloadAttachment?guid=950ab256-b485-4586-99b0-b27eb205fdfa`.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, *31*(3), 307–327. `https://doi.org/10.1016/0304-4076(86)90063-1`.

Boudoukh, J., Richardson, M., Stanton, R., & Whitelaw, R. F. (2004). MaxVaR: Long-horizon value at risk in a mark-to-market environment. *Journal of Investment Management*, *2*(3), 1–6. `https://pages.stern.nyu.edu/~rwhitela/papers/maxvar%20joim04.pdf`.

Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, *39*(4), 841–862. `https://doi.org/10.2307/2527341`.

Commodity Futures Trading Commission (n.d.). *Traders in financial futures: Historical compressed reports*. https://www.cftc.gov/MarketReports/CommitmentsofTraders/HistoricalCompressed/index.htm. Retrieved October 24, 2026.

Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, *13*(3), 253–263. https://doi.org/10.1080/07350015.1995.10524599.

Duffie, D., Fleming, M. J., Keane, F. M., Nelson, C., Shachar, O., & Van Tassel, P. (2023). *Dealer capacity and U.S. Treasury market functionality* (Staff Report No. 1070, August; revised October). Federal Reserve Bank of New York Staff Reports. https://doi.org/10.59576/sr.1070.

Ehm, W., Gneiting, T., Jordan, A., & Krüger, F. (2016). Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *78*(3), 505–562. https://doi.org/10.1111/rssb.12154.

Engle, R. F., & Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, *22*(4), 367–381. https://doi.org/10.1198/073500104000000370.

Federal Reserve Bank of New York (n.d.). *Primary dealer statistics* [Data set]. Federal Reserve Bank of New York. https://www.newyorkfed.org/markets/counterparties/primary-dealers-statistics. Retrieved October 23, 2026.

Giacomini, R., & White, H. (2006). Tests of conditional predictive ability. *Econometrica*, *74*(6), 1545–1578. https://doi.org/10.1111/j.1468-0262.2006.00718.x.

Gibbs, I., & Candès, E. J. (2021). Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems*, *34*, 1660–1672. https://proceedings.neurips.cc/paper_files/paper/2021/hash/0d441de75945e5acbc865406fc9a2559-Abstract.html.

Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, *106*(494), 746–762. https://doi.org/10.1198/jasa.2011.r10138.

Hansen, B. E. (1992). Testing for parameter instability in linear models. *Journal of Policy Modeling*, *14*(4), 517–533. https://doi.org/10.1016/0161-8938(92)90019-9.

Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, *79*(2), 453–497. https://doi.org/10.3982/ECTA5771.

Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, *13*(2), 281–291. https://doi.org/10.1016/s0169-2070(96)00719-4.

Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer. https://doi.org/10.1007/978-0-387-84858-7.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, *6*(2), 65–70. https://www.jstor.org/stable/4615733.

Hull, J., & White, A. (1998). Incorporating volatility updating into the historical simulation method for value-at-risk. *The Journal of Risk*, *1*(1), 5–19. https://doi.org/10.21314/jor.1998.001.

Koenker, R., & Bassett, G., Jr. (1978). Regression quantiles. *Econometrica*, *46*(1), 33–50. https://doi.org/10.2307/1913643.

Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *The Journal of Derivatives*, *3*(2), 73–84. https://doi.org/10.3905/jod.1995.407942.

Lindsay, D. (2020). *Net position time series dataset* (Version 2) [Data set]. Kaggle. https://www.kaggle.com/datasets/davidlindsay1979/net-position-time-series-dataset.

McNeil, A. J., & Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, *7*(3–4), 271–300. https://doi.org/10.1016/s0927-5398(00)00012-8.

Meinshausen, N. (2006). Quantile regression forests. *Journal of Machine Learning Research*, *7*(35), 983–999. https://www.jmlr.org/papers/v7/meinshausen06a.html.

Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, *55*(3), 703–708. https://doi.org/10.2307/1913610.

Petrunia, R., Sabetti, L., & Voia, M. (2019). *Tail risk in the automated clearing settlement system (ACSS)* (Discussion Paper No. 7). Payments Canada. https://www.payments.ca/sites/default/files/tail_risk_in_the_automated_clearing_settlement_system_acss.pdf.

Sussman, E. (2023). A strengthened Orlicz–Pettis theorem via Itô–Nisio. *Annals of Functional Analysis*, *14*(1), Article 22. https://doi.org/10.1007/s43034-022-00225-1.

van den Hoogen, S. (2017). *Intraday liquidity risk estimation using transaction data: An extreme value theory approach* [Master's thesis, Delft University of Technology]. https://resolver.tudelft.nl/uuid:1c0c4545-bb07-4916-aa37-1d4bea3b7872.

Wisniewski, W., Lindsay, D., & Lindsay, S. (2020). Application of conformal prediction interval estimations to market makers' net positions. In *Proceedings of the Ninth Symposium on Conformal and Probabilistic Prediction and Applications* (Proceedings of Machine Learning Research, Vol. 128, pp. 285–301). https://proceedings.mlr.press/v128/wisniewski20a.html.

Wisniewski, W. J. (2024). *Net position and retail trader behaviour: Implications for market makers' risk management strategies* [Doctoral thesis, Royal Holloway, University of London]. https://pure.royalholloway.ac.uk/en/publications/net-position-and-retail-trader-behaviour-implications-for-market-/.